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Consumer Search and Product Returns

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Abstract. We study a scenario in which a consumer has uncertain match with a product but can conduct costly prepurchase search to resolve this match. The firm selling the product jointly decides its price and return policy to maximally extract surplus by strategically managing who bears the cost of resolving the match, and when it is resolved. If the consumer's search cost is low relative to the firm's return processing cost, the firm induces the consumer to search and bear the cost of resolving the match before making the purchase decision. However, if search cost is high enough, the firm offers a return policy such that match is resolved after the purchase decision and the firm bears the processing cost of an unmatched returned product. By allowing search to be potentially inconclusive, we show that active search and returning may coexist and, in fact, be complementary such that offering a partial-refund return policy can encourage search and more efficient search can result in more returns due to a more lenient return policy. The firm finds it optimal to add restrictive clauses to the return policy along with offering free returns, and the presence of these clauses induces consumers to resolve match by conducting prepurchase search. In addition to providing novel insights into the interplay between search and returns, our model explains a wide range of pricing and return policies used by firms, and associated search and return behaviors of consumers.

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1. Introduction

Prior to purchasing a product, a consumer is often uncertain about whether they will like the product. Sellers offer return policies so that consumers can return products that they realize, after purchasing them, do not match their needs (for instance, the retailer Zappos that sells shoes online is well known for its liberal return policy[1]). Effectively, by offering returns, a retailer motivates postponing the determination of product match until *after* the purchase. Industry estimates suggest that, with more purchasing moving online and to formats like livestreaming, in the United States, approximately 10% of products (by dollar value) are returned, and these returns can be extremely costly for retailers.[2]

In many situations, however, consumers are able to search for information about a product before purchasing it. The information that a consumer searches may be obtained at the seller's website and store, through online consumer reviews, through conversations with other consumers, by inspecting the product, and so on. Some retailers go to extensive lengths to enable consumers to determine their match with a product (for instance, the retailer Warby Parker that sells eyeglasses online has a free prepurchase "Home Try On" program[3]). The search exercise, although costly to the consumer in terms of time and effort, may help the consumer in determining whether the product will match their needs *before* making the purchase decision.

In this paper, one of the key questions that we address is as follows. When and why would a retailer prefer these different approaches to product match determination, that is, match determination after purchase (when a return policy is offered, or even not offered) or before purchase (by motivating consumer search)? We build a game theory model in which a consumer can search to determine match with a product and the firm selling the product decides its price and the return policy. We start with a base model in which consumer search fully reveals match information, and find that the strategy that the firm follows depends primarily on the relative magnitudes of the consumer's cost of conducting search and the firm's cost of processing returns, as follows.

- If the firm's cost of processing returns is low relative to the consumer's cost of conducting search, then

the firm offers the product at a high price with free returns, that is, follows the strategy of match determination after purchase. In this case, the consumer does not search before purchasing; however, after purchasing (when match is assumed to be perfectly determined) if the product does not match their needs then they return it for a full refund. In this sense, the firm bears the cost of determining a match as it incurs the return processing cost.

- If the firm's cost of processing returns is high relative to the consumer's search cost, then the firm transfers the cost of determining a match to the consumer by inducing the consumer to search. Specifically, the firm offers the product at an intermediate price without a return policy—the price is not too high because then the consumer will not incur the search cost as the product may not match (i.e., the price is sufficiently low to "reimburse" the consumer for expending the search cost), and the price is not too low because then the consumer will just buy the product without search even if it is not a match. In this case, the consumer searches and match is determined before making the purchase decision, and they purchase the product only if a match is found.
- If both the firm's return processing cost and the consumer's search cost are high, then the firm offers the product at a low price and without a return policy. In this case, the consumer simply purchases the product without search because the price is low. Match is determined after purchase but the product cannot be returned even if it is not a match (which is why the price charged is low).

Matthews and Persico (2007) provide a working paper that studies the impact of search versus returns on product allocation efficiency and social surplus in a monopolistic scenario (questions that we also address), and Janssen and Williams (2024) examine similar questions in a competitive market scenario by extending the framework of Matthews and Persico (2007). Although certain aspects of the setup and forces examined in these papers are similar to those in our base model, the focus and treatment in these papers is quite different from ours as we conduct and interpret the analysis from the point of view of a monopolistic firm's joint decisions on price and return policy.

Even more importantly, however, a key assumption in Matthews and Persico (2007), Janssen and Williams (2024), and in our base model is that searching before purchase perfectly resolves match, which restricts them to the outcome that consumer search and product returns cannot exist simultaneously. This is because if a consumer conducts a search, then they will purchase only if the product matches, which means that they will not need to return the product and the return policy is irrelevant. Therefore, as an important conceptual next step, in this paper, we extend the base model to assume that consumer search does not perfectly reveal match but is potentially inconclusive, that is, search reveals match information with some likelihood, and with the remaining likelihood it does not provide any additional information. Then, even if a consumer conducts a search, if the search does not resolve match, the consumer may purchase the product and want to return it if after purchasing they realize that it does not match. Therefore, the return policy is important even though the consumer searches before purchase; in other words, with potentially inconclusive search, active search and returns can coexist. This aligns with empirical observations that consumers may have spent search effort before purchasing but may still return the product (Bechwati and Siegal 2005, Sahoo et al. 2018, Zhang et al. 2022, Ibragimov et al. 2024).

The conceptual addition of making search potentially inconclusive leads to rich additional insights. A novel insight is that search and returns cannot only coexist but, in fact, may be complementary, which means that offering a return policy can encourage search and that more informative search can lead to more returns. This is in stark contrast to the insight in Matthews and Persico (2007) and Janssen and Williams (2024) (and in our base model) that search and return are always substitutes.

We now provide the intuition behind the complementarity of search and returns with potentially inconclusive search. Specifically, consider the result that offering a return policy, with a partial refund (which only arises with potentially inconclusive search), can encourage search rather than discourage it. This happens if search cost is high enough. First, note that without a return policy, the firm does not want to motivate search when search cost is high because the price reduction needed to motivate search would be too large, and the firm prices the product low to motivate the consumer to buy without search. However, with a partial-refund return policy, the firm can charge a high price because the consumer keeps the product only if there is a match. Because the price is high and the refund for a return is only partial, the consumer now searches so that if search reveals that the product does not match they do not purchase it; if search reveals that the product matches, they purchase it and keep it; and if search reveals no new information then the consumer still purchases the product and if they find after purchase that it does not match, then they return it for a partial refund. In other words, the return policy increases the value of search because it allows the consumer to be partially protected from the downside while leveraging the upside of doing a search and obtaining information.

We extend the model with inconclusive search to incorporate hassle costs and complex return policies with restrictive return clauses (e.g., original packing

required for returning, 14-day return window, etc.). Here, we find that more efficient consumer search can result in more lenient return clauses, thereby leading to more returns. Moreover, we find that there may be costly consumer search even if returns are completely free. The reason is that both hassle costs and return policy clauses make the option of returning less attractive to consumers, and for them to value it sufficiently, the firm must make the return policy more attractive by making returns free. Therefore, generous return policies, such as those with large or free refunds, can arise even in a monopolistic situation and may reduce consumer surplus while increasing firm profit.[4] This goes against lay intuition that might rush to the interpretation that generous return policies benefit consumers while hurting firms (as expressed in certain practitioner articles[5]) and are usually a competitive response by firms. We also extend the model with inconclusive search to allow the firm to influence the informativeness of search and find that it will set the informativeness to be either the highest possible or the lowest possible. Essentially, the firm wants to either minimize search effectiveness for the consumer or minimize interim consumer heterogeneity due to potentially inconclusive search because interim consumer heterogeneity reduces its ability to extract consumer surplus. These insights are new and not available in previous literature.

Our research is related to the literature on consumer return policies (Moorthy and Srinivasan 1995; Che 1996; Shieh 1996; Davis et al. 1998; Anderson et al. 2009; Petersen and Kumar 2009, 2015; Shulman et al. 2009, 2010, 2011; Ofek et al. 2011; Hsiao and Chen 2012; Janakiraman et al. 2016). A main theme that runs across these papers is that returns have option value for consumers as they provide consumers the ability to return a product after purchasing, and many of these papers also include a hassle cost of returning a product. The papers listed above, however, do not connect return policy with consumer search before purchase, which is our focus. The existing literature on returns has identified various reasons to explain partial refunds for returns, including consumer opportunism (Chu et al. 1998), heterogeneous valuation uncertainty of consumers (Courty and Li 2000, Shulman et al. 2009), capacity (Xie and Gerstner 2007), demand uncertainty (Su 2009), competitive dynamics (Guo 2009, Shulman et al. 2011), product line design (Huang and Zhang 2020, Zou et al. 2020), and omnichannel considerations (Nageswaran et al. 2020). As mentioned earlier, we show that a reason for offering a partial refund for a return is to induce consumer search when it is potentially inconclusive, which is a new contribution to this literature.

There is also a large literature on consumer search; we refer the reader to Honka et al. (2019) for a comprehensive review of this literature. We note that search in our setting should be interpreted as information acquisition about one product rather than information acquisition about multiple products. Previous work in a single-product setting has used the "search" terminology as well (Branco et al. 2012, 2016). Ning and Villas-Boas (2023) study how a consumer balances gaining information about a product before and after purchase. This is related to our framing of search and return policies as determining match before and after purchase. At a high level, our paper explores firm strategies to affect information acquisition by consumers where consumers can do partial search (i.e., not obtain full information from search); these aspects are related to Dukes and Liu (2016) and Jerath and Ren (2021).

Other papers that jointly study consumer information acquisition and return policies are Petrikaite (2018) and Hinnosaar and Kawai (2020). However, unlike our formulation, in these papers, consumers do not actively do prepurchase information acquisition. In Petrikaite (2018), consumers purchase competing products sequentially to find a match and can return a product if there is no match and go on to the next product—There is no prepurchase information acquisition. In Hinnosaar and Kawai (2020), consumers are exogenously differently informed before purchase about their product valuations and do not make a decision to search or not, whereas they can return the product if they do not find a match after purchasing; furthermore, in this paper, the firm maximizes guaranteed profit (robust optimization). Choi et al. (2024) study an emerging ship-then-shop retail strategy in which a firm's prediction algorithm substitutes consumer search and ships products to consumers under a free return policy.

There is also a large literature on the idea that the firm may strategically prevent or encourage information acquisition by consumers (Crémer and Khalil 1992, Lewis and Sappington 1997, Crémer et al. 1998, Bergemann and Valimaki 2002, Dukes and Liu 2016, Jerath and Ren 2021). Adding to this literature, we show that the return policy can also be a profitable tool for the firm to prevent or encourage information search by consumers, and we show how the design of return policy varies with the consumer search characteristics such as search cost and search effectiveness.

The rest of the paper is organized as follows. In Section 2, we describe our base model where search fully resolves match uncertainty, and discuss the key results and insights that obtain from it. In Section 3, we incorporate potentially inconclusive search into the base model and obtain further insights, including through considering various extensions. In Section 4, we conclude with a discussion. We provide proofs of all our results in the appendix.

2. Base Model and Results

We develop a model with one firm selling one product and one consumer who buys at most one unit of the

product. The product may or may not match the consumer's needs. If the product matches the consumer's needs, then they obtain consumption utility of one, and if the product does not match their needs, then they obtain consumption utility of w, where $0 < w < 1$. Prior to purchase, the consumer knows the product price $0 \le p \le 1$ but is uncertain about the match with the product. We assume that the product match is always fully resolved after purchase. The prior probability that the product matches the consumer's needs is given by $0 \le q \le 1$, and this is shared as prior belief by the consumer and the firm.

If the consumer purchases the product and the product matches their needs, then they obtain net utility $1-p$. If the consumer purchases the product and it does not match, then if the firm does not allow product return, the consumer cannot return the ill-matched product and thus obtains utility $w-p$; instead, if the firm allows the product return, then the consumer can return the ill-matched product and obtains net utility $-\gamma$, where γ is the return fee set by the firm (note that the consumer would receive a refund of $p-\gamma$). We assume away a hassle cost of returning the product (in an extension, we analyze how a hassle cost affects equilibrium outcomes). We assume that the utility of the consumer's outside option is 0. For easy exposition, we denote a binary variable $\theta \in \{0,1\}$ as the firm's return policy. Specifically, if $\theta = 1$, it means that the firm allows product return; if $\theta = 0$, it means that the firm does not allow product return.

If the consumer returns a product, the firm incurs a return processing cost $\xi > 0$. After incurring this cost, the firm can sell the product to a different consumer (this can be thought of as the firm "salvaging" the returned product). For simplicity, we assume that consumers do not associate any reduced valuation with such products, that is, a matched product still gives consumption utility of one and an ill-matched product gives consumption utility of w. Allowing for a reduced valuation of buying previously returned products will have no qualitative impact on our results and insights, but complicates the analysis.[6] We assume the firm's marginal cost of production of the product is $\psi \ge 0$.

Prior to purchase, the consumer can conduct search that reveals information to reduce the uncertainty about product fit by incurring a search cost $c > 0$. In the base model, we assume that match uncertainty is fully resolved if the consumer conducts the search. Therefore, the consumer who searches would find, before purchase, a match with probability q and find no match with probability $1-q$, and make her decision accordingly.

The timing of the game is the following. First, the firm decides the price p and whether to allow product return (i.e., either $\theta = 1$ or $\theta = 0$); if the firm allows product return, it also sets the return fee γ. Next, the consumer decides whether to search. If the consumer searches, they will buy if they find a match and not buy otherwise. If the consumer buys without search and the firm allows product return, then the consumer decides whether to return the product by paying the return fee γ to the firm. We solve for the subgame-perfect equilibrium of the game. We discuss the analysis and results for this model below.

First, given the firm's decisions on p, θ and γ, we analyze the consumer search and purchase decision. If the consumer buys the product without search, then with probability q the purchased product matches the consumer's needs and thus the consumer gets utility $1-p$, whereas with probability $1-q$ the purchased product does not match, and the consumer gets utility $w-p$ if $\theta = 0$ or gets utility $-\gamma$ by returning the product if $\theta = 1$. Therefore, the expected consumption utility based on prior belief, denoted by EU_{prior}, is given by

$$EU_{\text{prior}} = \underbrace{q(1-p)}_{\text{match}} + \underbrace{(1-q)(\overbrace{(1-\theta)(w-p)}^{\text{return not allowed}} + \overbrace{\theta(-\gamma)}^{\text{return allowed}})}_{\text{no match}}. \quad (1)$$

Without search, the consumer would buy the product if and only if $EU_{\text{prior}} \ge 0$. Therefore, the expected utility of not searching, denoted by EU_{ns}, is given by $EU_{\text{ns}} \equiv \max\{EU_{\text{prior}}, 0\}$.

If the consumer searches before purchase, with probability q the consumer finds a match and buys the product, and with probability $1-q$ the consumer does not find a match and thus does not buy the product and obtains zero utility. Therefore, the expected utility of search, denoted by EU_{s}, is given by

$$EU_{\text{s}} = q(1-p) + (1-q)0 - c.$$

Note that the consumer will purchase the product in the following scenarios:

1. If $EU_{\text{s}} \ge EU_{\text{ns}}$, the consumer will search; match is resolved before the purchase decision is made, and the consumer will buy the product only if there is a match.
2. If $EU_{\text{s}} < EU_{\text{ns}}$ and $EU_{\text{prior}} \ge 0$, the consumer will buy the product without search, and match is resolved after purchase.

Next, we discuss the firm's optimal strategies to induce the two purchase scenarios above. For the first purchase scenario where the consumer searches product information and buys the product only if they find a match through search, the firm can obtain profit $E\pi_{(1)} = q(p-\psi)$ (the firm only needs to produce q products), and thus it solves the following profit maximization problem:

$$E\pi^*_{(1)} = \max_p \; q(p-\psi)$$
$$\text{s.t. } EU_{\text{s}} \ge EU_{\text{ns}}.$$

For the second purchase scenario where the consumer buys without any search, with probability $1-q$, the consumer will find that the product does not match, and they can return the ill-matched product if the firm allows product returns. Therefore, the firm solves the following profit maximization problem:

$$E\pi^*_{(2)} = \max_{p,\theta,\gamma} \; qp + (1-q)((1-\theta)p + \theta(\gamma - \xi)) - (\theta q + (1-\theta))\psi$$
$$\text{s.t.} \quad EU_s < EU_{ns} \text{ and } EU_{prior} \geq 0 \text{ and } 0 \leq \theta\gamma \leq \theta(p-w).$$

We provide the details of the rest of the solution in Section A.1 in the appendix. In Table 1, we show the equilibrium outcomes, and in Figure 1, we show the regions where the different equilibrium outcomes obtain. We now discuss these outcomes.

Region A in Figure 1 comprises of Regions 1(b) and 2(b) in Table 1. Here, the return processing cost is not high, whereas the search cost is relatively high. Therefore, the firm allows product returns to induce the consumer to purchase without search, and match uncertainty is resolved after purchase. The firm charges a high price to extract full surplus from the consumer in case of a match, whereas if a match is not found and the ex post valuation is low, then the consumer can return the product for free.

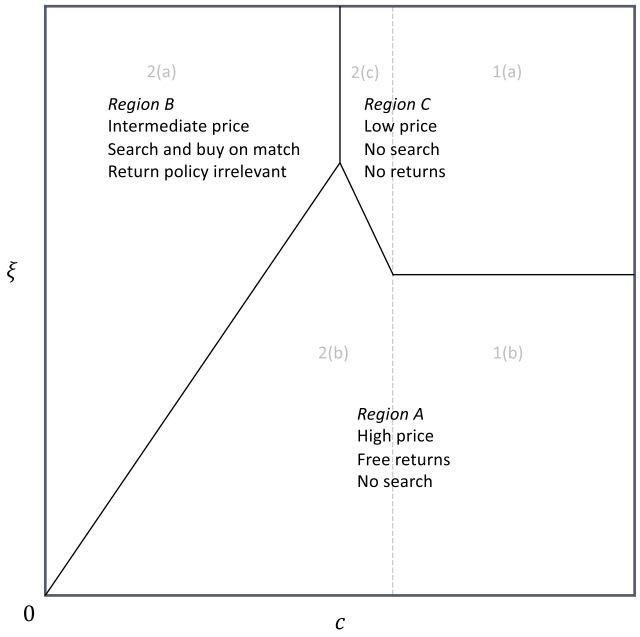
Region B in Figure 1 comprises of Region 2(a) in Table 1. Here, the search cost is low and return processing cost is high. The firm induces the consumer to search and invest in resolving the uncertainty before purchase because returns are costly for the firm. Because the consumer will buy only if a match is found after search, there will be no returns and the return policy is irrelevant. However, to induce search from the consumer, the firm has to reduce the price of the product because there is a chance that the consumer expends the search cost but does not find a match; that is, the price is sufficiently low to "reimburse" the consumer for expending the search cost. The price, however, cannot be too low because then the consumer will buy without searching.

Region C in Figure 1 comprises of Regions 1(a) and 2(c) in Table 1. Here, both the search cost and the return processing cost are high. The firm charges a very low price to induce the consumer to buy without search; that is, the price is sufficiently low such that the consumer does not feel the need to determine a match before purchasing even though there is a chance that the product will not match. The firm also does not allow product returns because the low price cannot support returns; the consumer still buys because the price is low enough. It is worth noting that in Region 2(c), the price is so low that the consumer can obtain a positive expected surplus (in all other regions, the surplus is zero). The following proposition states the key insights.

Table 1. Equilibrium Outcomes for the Base Model

Region in Figure 1 and parametric condition	Firm strategy and induced consumer response	Expected firm profit and expected consumer surplus
Region 1(a) Return processing cost is high and search cost is sufficiently high ($c \geq q(1-q)(1-w)$ and $\xi > \psi - w$).	Firm does not accept product returns Induces consumer to buy without search Charges a low price ($p^* = q + (1-q)w$).	$q + (1-q)w - \psi$ and 0
Region 1(b) Return processing cost is low and search cost is sufficiently high ($c \geq q(1-q)(1-w)$ and $\xi \leq \psi - w$).	Firm offers returns without charging any return fee Induces the consumer to buy without search Charges the highest-possible price ($p^* = 1$).	$q(1-\psi) - (1-q)\xi$ and 0
Region 2(a) Return processing cost is high while search cost is low ($c < q(1-q)(1-w)$ and $\xi \geq \frac{c}{1-q}$ and $c(\frac{1}{1-q}+1) \leq q(1-\psi) + \psi - w$).	Firm induces the consumer to search and to buy only if they find a match Charges an intermediate price ($p^* = 1 - \frac{c}{q}$; note that $p^* > q + (1-q)w$) There is no product return.	$q(1-\psi) - c$ and 0
Region 2(b) Return processing cost is not too high while search cost is medium ($c < q(1-q)(1-w)$ and $\xi < \frac{c}{1-q}$ and $q(1-\psi) - (1-q)\xi \geq \frac{c}{1-q} + w - \psi$).	Firm offers returns without charging any return fee Induces the consumer to buy without search Charges the highest-possible price ($p^* = 1$).	$q(1-\psi) - (1-q)\xi$ and 0
Region 2(c) Return processing cost is high while search cost is medium ($c < q(1-q)(1-w)$ and $c(\frac{1}{1-q}+1) > q(1-\psi) + \psi - w$ and $q(1-\psi) - (1-q)\xi < \frac{c}{1-q} + w - \psi$).	Firm does not allow product returns; Induces the consumer to buy without search; Charges a very low price ($p^* = \frac{c}{1-q} + w$; note that $p^* < q + (1-q)w$).	$\frac{c}{1-q} + w - \psi$ and $q(1-w) - \frac{c}{1-q} > 0$

Figure 1. Equilibrium Outcomes in the Base Model w.r.t. Search Cost (c) and Return Processing Cost (ξ) ($w = 0.2$; $q = 0.5$; $\psi = 0.4$)



Proposition 1. *The following outcomes obtain the following:*

1. *When the consumer search cost is relatively high compared with the return processing cost, the firm offers the product at a high price with free returns. The consumer purchases the product without search and returns the product if a match is not realized after purchase.*

2. *When the return processing cost is relatively high compared with the consumer search cost, the firm offers the product at an intermediate price and induces the consumer to search before purchasing. The consumer purchases only if a match is found and the return policy is irrelevant.*

3. *If both the return processing cost and the consumer search cost are high, the firm offers the product at a low price without a return policy. The consumer purchases the product without searching and is unable to return it even if it does not match.*

The consumer surplus is zero except when the return processing cost is high and the search cost is intermediate.

From a practical viewpoint, Region A corresponds to the strategy that the retailer Zappos uses, whereas Region B corresponds to the strategy that the retailer Warby Parker uses. One can reasonably argue that the return processing cost for a pair of prescription glasses before they can be resold is high for Warby Parker (e.g., the prescription lenses go to waste and only the frame can be salvaged and resold), whereas the return processing cost for a pair of shoes before they can be resold may not be very high in comparison; on the other hand, Warby Parker's Home Try On program reduces search cost. Region C corresponds to retailers for whom both the return processing cost and the consumer search cost are high compared with the value of the product, such that returns are not offered and consumers also do not expend much effort in search. Examples of such retailers could be stores that sell low-value products and typically do not offer return policies, such as Dollar Tree.[7]

We also obtain interesting comparative statics on quantities such as price, existence of return policy, firm profit and consumer surplus with respect to the search cost, c, the return processing cost, ξ, the marginal production cost, ψ, and the prior, q. We do not go into details here, but these can be determined from the expressions in Table 1. Next, to distill out more insights, we compare the above outcomes with outcomes in two benchmark scenarios: one in which search is exogenously shut down and one in which returns are exogenously shut down.

First, to crystallize the role of consumer search, we compare the solution of the base model with a benchmark scenario in which we exogenously shut down consumer search. The solution of this scenario corresponds to the solution of the base model for large enough search cost c, specifically, $c > q(1-q)(1-w)$. This solution is the following: if the return processing cost is high enough, the firm does not allow product returns and induces the consumer to purchase by charging a low price; otherwise, the firm charges a high price and induces the consumer to purchase by allowing free product returns. We obtain several interesting insights, as presented in the following corollary. A detailed treatment is provided in Section A.1.1 in the appendix.

Corollary 1. *When the consumer can search, compared with a benchmark without search:*

1. *Consumer search weakly reduces firm profit, except if the marginal production cost and return processing cost are relatively high while the search cost is small, in which case consumer search increases firm profit.*

2. *Depending on the search cost and return processing cost, consumer search can deter the firm from offering returns or encourage the firm to allow returns.*

In general, as one would expect, the consumer's ability to search to obtain information before making the purchase decision helps the consumer and hurts the firm. However, under the conditions in Corollary 1(1), consumer search can increase the firm's profit. This is because under a high marginal production cost and high return processing cost, the firm would like to maintain a high price and also not have returns—A low search cost enables precisely this because the consumer does not have to be "compensated" much to motivate a search so that price can be high, whereas the search fully resolves match and a consumer only buys under match and therefore there are no product

returns. (This happens in Regions I and II in Figure A.2 in the appendix.)

Regarding Corollary 1(2), following the logic above, consumer search can remove the need to offer returns because search resolves match before purchase. (This happens in Region I in Figure A.2 in the appendix.) However, under some circumstances, the possibility of consumer search can motivate a return policy, resulting in returns (compared with when a return policy would not be offered without the possibility of search), which is counterintuitive precisely because search resolves match before purchase. This happens when the search cost is relatively high, but the return processing cost is intermediate (specifically, Region IV in Figure A.2 in the appendix). In the model with search, the firm offers a high price and free returns to induce the consumer to purchase without searching, whereas in the benchmark without search the firm offers the product at a low price and no returns. Overall, consumer search can reduce returns or motivate offering a return policy, depending on the costs of search and processing returns. Generally speaking, consumer search reduces firm profit even if it reduces returns (Shulman et al. (2009) have a similar result, although through a different mechanism).

Second, to crystallize the role of a return policy, we compare the solution of the base model with a benchmark scenario in which we exogenously shut down returns. The solution of this case corresponds to the solution of the full model for large enough return processing cost ξ, specifically, $\xi > \frac{q(1-\psi)+\psi-w}{2-q}$. Allowing returns can save the consumer's search cost because the consumer can wait to resolve match uncertainty until after purchase. However, it is the firm that decides whether to allow a return or not, which gives the firm an advantage. The key insight we obtain is provided in the following corollary. A detailed treatment is provided in Section A.1.2 in the appendix.

Corollary 2. *Compared with a benchmark without returns, allowing returns weakly benefits the firm but may hurt the consumer* (*specifically, under medium search cost*).

Allowing returns helps the firm because if search cost is high enough the firm can offer a return policy and postpone match determination until after purchase; therefore, the firm does not have to "compensate" the consumer for the cost they incur to conduct a search to determine match and can charge a high price. Because the firm can charge a high price, this can hurt the consumer (which happens, specifically, in Region B in Figure A.4 in the appendix).

Overall, our base model highlights the roles of and the interplay between return policies and consumer search and reveals when and why each should be offered. Based on the relative magnitudes of the cost of processing returns and the cost of the consumer conducting a search, the firm makes a tradeoff about who should bear the responsibility of determining match, the firm (through a return policy) or the consumer (through consumer search). A key takeaway is that consumer search can be used to reduce returns and save return processing costs for the firm when this cost is high. On the other hand, return policy can be employed to offset the negative impact of consumer search on the firm's pricing capability when the search cost is high. In general, search benefits the consumer and can benefit or hurt the firm, whereas return policy benefits the firm and hurts the consumer.

3. Model with Inconclusive Search

In the base model, we assume that the search can fully resolve match uncertainty. Therefore, in the equilibrium where the consumer chooses to search, there would be no return policy. In other words, consumer search and product returns do not coexist in equilibrium. However, we observe in reality that consumers in many cases conduct search to obtain information about the product before purchasing it even when a firm offers a return policy (Bechwati and Siegal 2005, Sahoo et al. 2018, Zhang et al. 2022, Ibragimov et al. 2024). Furthermore, Ibragimov et al. (2024) and Zhang et al. (2022) document that consumers often return products even after searching before purchasing. This implies that prepurchase search does not always fully resolve product valuation for the consumer as effectively as experiencing the product postpurchase can.[8]

In this section, we incorporate potentially inconclusive search into the base model. With everything else being the same as the base model, we assume that with probability δ the search is conclusive in the sense that the consumer would perfectly know the product match (the consumer would find a match with probability q and not find a match with probability $1-q$), and with probability $1-\delta$ the search is inconclusive in the sense that the consumer gets no new information through search and thus cannot update their belief in this case. We call $0 \le \delta \le 1$ the "search informativeness."[9] It is possible that consumers may not be able to find out the product information that matters to them after conducting the search. For example, suppose that a consumer is looking for product information on a product reviews website. Although there may be hundreds of product reviews for the product, only a few of them may mention the attributes that the consumer cares about. Therefore, after conducting a search on dozens of reviews, the consumer may still not be able to obtain the information they are looking for (and they have limited time and thus cannot read all of the available reviews).[10]

As in the base model, the expected utility based on prior belief, EU_{prior}, is given by (1) and $EU_{\text{ns}} \equiv \max$

$\{EU_{\text{prior}}, 0\}$. The expected utility of searching before purchase is given by

$$EU_s = \underbrace{\delta(q(1-p)+(1-q)0)}_{\text{conclusive search}} + \underbrace{(1-\delta)EU_{ns}}_{\text{inconclusive search}} - c. \quad (2)$$

Note that the consumer will purchase the product in the following three scenarios:

1. If $EU_s \geq EU_{ns}$ and $EU_{\text{prior}} < 0$, the consumer would search and buy the product only if they find a match through search. In this case, there is no need to return product.
2. If $EU_s \geq EU_{ns}$ and $EU_{\text{prior}} \geq 0$, the consumer would search and buy the product if they find a match through search or if the search is inconclusive (because $EU_{\text{prior}} \geq 0$). In this case, the consumer can return the ill-matched product if the firm allows product return.
3. If $EU_s < EU_{ns}$ and $EU_{\text{prior}} \geq 0$, the consumer would buy the product without search.

The second scenario above, in which search and returns can coexist, is the new scenario that is generated here as compared with the base model.

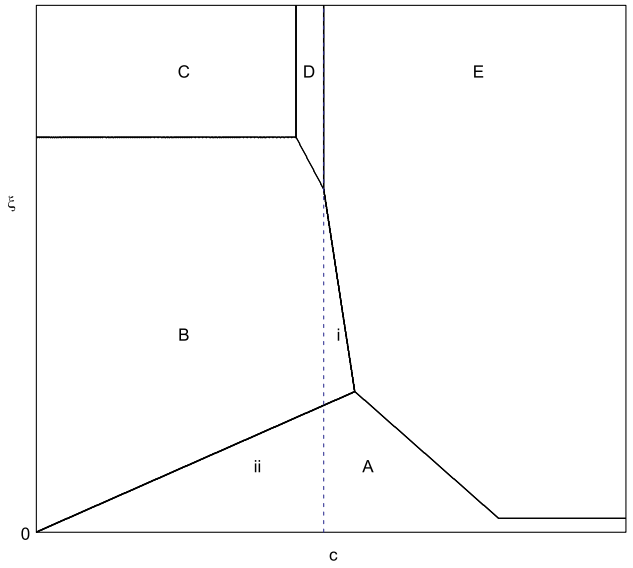
Note that with inconclusive search interim consumer heterogeneity arises even though there is no ex ante consumer heterogeneity. In other words, before a search, a consumer believes that the product matches with probability q. However, after a search, with probability $1-\delta$, the consumer will still believe that the product matches with probability q, but with probability δq the consumer will have found a match and with probability $\delta(1-q)$ the consumer will not have found a match. (In the base model, because $\delta = 1$, only the last two outcomes exist.)

To solve for the equilibrium outcomes, we first derive the firm's optimal strategies to induce the three purchase scenarios above. Then, we compare the profits of different strategies to determine which one the firm will follow under different parametric conditions. We show these details in Section A.2 in the appendix. The equilibrium outcomes are specified in Proposition 2 and illustrated in Figure 2.

Proposition 2. *When inconclusive search is considered, the following holds:*

1. *If return processing cost is low and search cost is neither too low nor too high, the firm offers a generous return policy without charging any return fee (i.e., $\theta^* = 1, \gamma^* = 0$) to induce the consumer to buy without search, charges the highest price (i.e., $p^* = 1$), and obtains expected profit $E\pi^* = q(1-\psi) - (1-q)\xi$. The consumer surplus is zero. This corresponds to Region A in* Figure 2.
2. *If search cost is small and the return processing cost is intermediate, the firm induces the consumer to search and to buy the product if they find a match or if the search is inconclusive; it charges a relatively high price $p^* = 1 - \frac{c}{\delta q}$, allows product return (i.e., $\theta^* = 1$) with a return fee $\gamma^* = \frac{c}{\delta(1-q)}$, and obtains expected profit $E\pi^* = q - c - (1-\delta)(1-q)\xi - q\psi$. The consumer surplus is zero. This corresponds to Region B in Figure 2.*
3. *If the search cost is small and the return processing cost is high, the firm induces the consumer to search and to buy only if they find a match through search; it charges a relatively high price $p^* = 1 - \frac{c}{\delta q}$, there is no product return, and obtains expected profit $E\pi^* = \delta q(1-\psi) - c$. The consumer surplus is zero. This corresponds to Region C in Figure 2.*
4. *If the search cost is intermediate and the return processing cost is high, the firm induces the consumer to search and to buy if they find a match or if the search happens to be inconclusive; it charges a low price $p^* = q + (1-q)w$, does not allow product return, that is, $\theta^* = 0$, and obtains expected profit $E\pi^* = (\delta q + 1 - \delta)(q + (1-q)w - \psi)$. The consumer surplus is positive and is given by $\delta q(1-q)(1-w) - c$. This corresponds to Region D in Figure 2.*
5. *If the search cost is high and the return processing cost is high, the firm does not accept product returns ($\theta^* = 0$), charges a low price (i.e., $p^* = \min\{q + (1-q)w, \frac{c}{\delta(1-q)} + w\}$) to induce the consumer to buy without search, and obtains expected profit $E\pi^* = \min\{q + (1-q)w - \psi, \frac{c}{\delta(1-q)} + w - \psi\}$. The consumer surplus is given by $\max\{0, q(1-w) - \frac{c}{\delta(1-q)}\}$. This corresponds to Region E in Figure 2.*

Figure 2. Equilibrium Outcomes with Potentially Inconclusive Search w.r.t. Search Cost (c) and Return Processing Cost (ξ) ($w = 0.38; q = 0.5; \delta = 0.81; \psi = 0.4$)



From the results, the primary novel outcome that is due to potentially inconclusive search is that search and returns can coexist and be complementary. We discuss this outcome here and other outcomes in Section A.2 in the appendix. The outcome of interest, which corresponds to Region B in Figure 2, happens when the return processing cost is intermediate and the search cost is low. Here, the firm offers a price that is not too low along with a return policy with a moderate return fee, and the consumer conducts the search and buys on

match and also on unsuccessful search. This corresponds to the practices for many retailers where the retailers price such that consumers search for match information and also accept product returns with a return fee.

The intuition is that consumer search can help resolve the match uncertainty, and thus under an intermediate return processing cost, the firm has the incentive to induce the consumer to search first and also offers returns if search fails to provide information. By doing so, the firm can still charge a relatively high price and also consumer search helps reduce the probability of return and thus reduce the expected return processing cost for the firm.

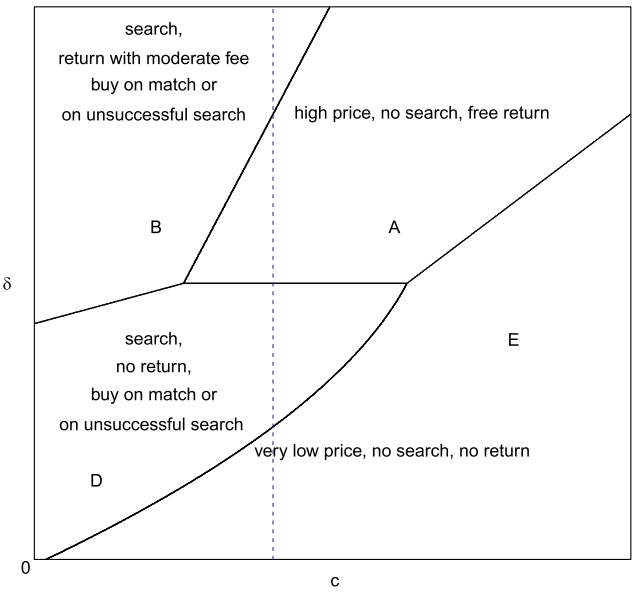
More interestingly, we also find that compared with the case where return policy is exogenously shut down (Figure A.5 in the appendix), offering a return policy can lead to consumer search when not offering a return policy would not lead to consumer search (as in the subregion i in Region B in Figure 2). This is counterintuitive as returns would generally be expected to have the effect of decreasing search (as in the subregion ii in Region A in Figure 2) or not affecting it. The intuition behind this finding is that with returns the consumer can buy and return the mismatched product even if their search fails to provide new information, and thus the consumer has higher incentive to search. This motivates the firm to promote more consumer search and charge a higher price (compared with the case without returns). Overall, offering a return policy can lead to consumer search; in this sense, search and product returns can be complementary, which is an important new insight to the literature. Moreover, this search-inducing effect of a return policy is strongest when δ is moderate (specifically, the area of subregion i is maximal for $\delta = \frac{\sqrt{1+4(1-q)^2}-1}{2(1-q)^2}$). This is because when δ is small consumers maintain reluctant to search even if return policy is offered, whereas when δ is large, consumers are already highly motivated to search even without a return policy.

It is also important to note that a moderate return fee, that is, a partial refund, is strategically employed to induce consumer to search in Region B in Figure 2. If the return fee is too low, the consumer will simply buy the product without any search and then the firm will receive a return with a high probability. If the return fee is too high, the consumer will then not buy the product when search fails. As noted earlier, the search-inducing role of a return policy through partial refunds is new to the literature, and this suggests that a firm can offer a return policy with partial refund to promote consumer search (which reduces the number of returns and increases the price). Furthermore, we find that a lower search cost can result in a more lenient return policy with a larger refund. This is because when the consumer is more willing to search, the firm offers a more generous return policy so that it can set a higher product price and the consumer still wants to search. We summarize the key novel insights obtained from the model with potentially inconclusive search in the following corollary.

Corollary 3. *When search can be inconclusive, if return processing cost is intermediate and search cost is small (Region B in Figure 2), consumer search and product returns can coexist and be complementary. The firm strategically offers a return policy with moderate return fee to induce consumer search, where the return fee decreases as search becomes more efficient. The search-inducing effect of a return policy is strongest when δ is moderate.*

Next, we investigate the roles played by the parameter δ, the informativeness of search when search can be inconclusive, and the parameter c, the cost of search. We do this with the aid of Figure 3, in which we plot c on the x-axis and δ on the y axis—Larger values on the x axis and smaller values on the y axis indicate less efficient search (roughly speaking, the information obtained from search relative to the cost of search). We can see that if the cost of search is small enough (c is small enough; Regions B and D) then the firm motivates search otherwise it does not, if informativeness of search, δ, is large enough (Regions A and B), then the firm offers a return policy; otherwise, it does not, and if c is large and δ is small (Region E), then there is no search or return. Therefore, although both these parameters are related to search efficiency, the roles they play are subtly different, as we highlight in the following corollary.

Figure 3. (Color online) Equilibrium Outcomes w.r.t. Search Cost (c) and Search Informativeness (δ) ($w = 0.2; q = 0.5; \xi = 0.1; \psi = 0$)



Corollary 4. *Broadly speaking, based on the search cost, c, the firm primarily decides whether to price low enough to motivate search or not, and based on the informativeness of the search, δ, the firm primarily decides whether to offer a return policy to prevent or induce search.*

Furthermore, Figure 3 also helps us to distill out other interesting comparative statics w.r.t. δ. When the firm induces the consumer to search, we find that higher search informativeness (i.e., larger δ) can hurt the firm. Specifically, in Region D, the firm induces the consumer to search and the firm charges a relatively low price (equal to the consumer's prior utility, given by $q+(1-q)w$) to induce the consumer to buy even if their search fails to provide new information, but they cannot return the mismatched product. In this case, for larger δ, there will be more informed consumers who would not purchase due to mismatch, which decreases the sales. However, the firm cannot adjust to a higher price with a larger δ because it has to maintain a low price to induce the uninformed consumers to buy. Overall, even if the optimal strategy for the firm is to induce consumer search, the firm's profit can decrease with search informativeness.

Following the above point, we also find that higher search informativeness does not necessarily induce the firm to promote consumer search. Specifically, as indicated by the dashed line in the Figure 3, when δ becomes larger (but not too large), and the regime switches from Region D to Region A, the firm switches from inducing search to preventing search through offering a generous return policy, because such a return policy allows the firm to charge a high price. In this case, consumer surplus reduces because consumers obtain a positive expected surplus in Region D but obtain zero expected surplus in Region A. When δ becomes large enough, and the regime switches from Region A to Region B, the firm switches from preventing search through a generous return policy to inducing search through a relatively strict return policy with moderate return fee. This is because a large enough δ can help the firm to significantly reduce the number of returns. In terms of the profit, based on Proposition 2, the firm's profit decreases with δ in Regions D and E, whereas it increases with δ in Regions B and C. Consumer surplus generally increases with δ; however, when δ reaches an intermediate level and shifts from Region D to Region A, expected consumer surplus drops from a positive value to zero.[11] We summarize the key results in the following corollary.

Corollary 5. *Higher expected informativeness of search, that is, a larger value of δ, may lead to more or less consumer search, higher or lower firm profit, and higher or lower consumer surplus, through its implications on the optimal price and return policy.*

In summary, incorporating potentially inconclusive search into the base model produces the key novel insight that search and returns can coexist and be complementary, and that the firm may only offer a partial refund on returns. Next, we extend the model with potentially inconclusive search: In Section 3.1, we incorporate a hassle cost of returning incurred by the consumer; in Section 3.2, we allow the firm to incorporate clauses into the return policy; and in Section 3.3, we allow the firm to endogenously set search informativeness. For ease of exposition, we assume for these extensions that the marginal production cost is zero, that is, $\psi=0$.

3.1. Hassle Cost

In this section, we extend the model by assuming that consumers incur a hassle cost $h>0$ to return the products. A hassle cost is different from a return fee because the consumer incurs it when returning a product but the firm does not benefit from it. We provide an outline of the solution of this model in Section A.2.1 in the appendix.

With a hassle cost, we find that our main insights still hold and some new insights about return policy arise. Specifically, there are scenarios in which the consumer is induced to search and yet the return policy is generous (no return fee, $\gamma^*=0$), an outcome that did not appear in the model without hassle cost (where if a consumer is induced to search and a return policy is offered, then the return fee is not zero). However, this is consistent with our observation that firms often charge no return fee but consumers still conduct search before purchase.

This outcome happens in Region B1 in Figure A.6 in the appendix. The intuition is that when the hassle cost is not low, the firm has to offer a generous return policy to induce the consumer to buy when their search fails. Interestingly, we find that in this case the consumer can also obtain a positive surplus and would benefit from higher hassle cost and from a lower chance of match (the expected consumer surplus is given by $EU^*=\delta(1-q)h-c$). The intuition is that as the hassle cost becomes higher or there is a lower chance of match, the firm has to charge a lower price to induce the consumer to search and purchase when their search fails (return fee is endogenously set to be zero). Therefore, the consumers can benefit from higher hassle cost and lower chance of match. This also explains why sometimes a firm has the incentive to reduce the consumer's hassle cost (e.g., offering return shipping label). We summarize the discussion above in the following proposition.

Proposition 3. *When c is small and h is moderate, expected consumer surplus increases in the hassle cost of*

returning. This happens in the equilibrium where the firm offers a generous return policy (i.e., with a full refund) and the consumer is induced to search and buy even if the search fails.

3.2. Complexity of Return Policy Through Return Clauses

It is common that a firm may accept a product return only under certain conditions (e.g., "must return within 14 days after purchase," "must return in the original packaging," "must have the original receipt," etc.). We call such conditions as the "complexity" of the return policy. The more complex the return policy is (i.e., the more conditions the return policy has), the more difficult it is for the consumer to return an ill-matched product. Note that this is different from a hassle cost since complexity affects the eligibility of a return. Therefore, the question arises that when does the firm have the incentive to complicate the return contract, and to what degree.

In this section, we extend the model to allow the firm to choose the complexity of the return policy along with other decisions in the base model. Specifically, we allow the firm to choose any value $\theta \in [0,1]$ (in our base model, the firm can only choose either $\theta = 1$ or $\theta = 0$); θ can be perceived as the probability of the consumer being able to return the ill-matched product. A more complex return policy implies a lower θ. For instance, the more complex the return policy is, i.e., the more conditions it has, the more likely it is that the consumer randomly violates the policy (e.g., the consumer randomly misplaces the receipt or loses the original packaging).

We provide an outline of the solution of this model in Section A.2.2 in the appendix and obtain the equilibrium outcomes as shown in Figure A.7. An interesting new finding is that when the firm offers a return policy, it makes the return policy complex to some degree (i.e., $0 < \theta^* = 1 - \frac{q}{1-q}\frac{c}{\delta q(1-w)-c} < 1$) so that there is a positive probability, given by $1-\theta^*$, that the consumer is not able to return the ill-matched product. However, at the same time, the firm offers a full refund (i.e., $\gamma^* = 0$) to the consumer if the consumer is able to return the ill-matched product (which occurs with probability θ^*).

The intuition is that the firm always prefers no return to a return because a returned product incurs a processing cost. Therefore, the firm increases the complexity of the return policy and thus reduces the chance of returning as much as possible. However, to achieve this, the firm has to decrease the return fee to maintain the attractiveness of the return policy to the consumer. Eventually, the firm keeps increasing the complexity of return policy until the commensurate return fee becomes zero.

It is also worth noting that the complexity of the return policy decreases as search becomes more efficient for the consumer (i.e., θ^* decreases in c and increases in δ). In other words, a cheaper or more informative search can lead to more lenient return clauses, thereby increasing returns. The intuition is that the firm designs a return policy such that the consumer is indifferent between search and buying without search (and the consumer chooses to search). Therefore, if search becomes more efficient for the consumer, the firm has the incentive to decrease the complexity of the return policy so that the firm can charge a higher product price and the consumer will still be indifferent and choose to search. We summarize the discussion above in the following proposition.

Proposition 4. *If the firm offers a return policy, it is characterized by some complexity (i.e., $0 < \theta^* < 1$) and a full refund (i.e., $\gamma^* = 0$). The consumer searches even though the return policy offers a full refund. A more efficient search (lower c and/or higher δ) can lead to a less complex return policy, thereby increasing expected returns.*

The main takeaway from this extension is that the firm finds it optimal to offer a return policy with a full refund if it can add an appropriate level of complexity to it (by adding an appropriate number of clauses to the policy that the consumer may not be able to always meet, e.g., original packaging, original receipt, limited return window, etc.). We note that this aligns well with market practice where companies often offer "free returns" but various clauses are added to the return policy.

3.3. Endogenous Search Informativeness

Under a return policy, does the firm have the incentive to increase or decrease the search informativeness (i.e., δ)? Specifically, the firm can typically decide how much information to provide on its official web pages or in its ads. The more information is provided by the firm, the more likely it is for a consumer to find useful information during search to update their belief. It is natural to ask how much information the firm should provide through a consumer search, and how this interacts with the return policy.[12]

To answer this question, we extend the model by allowing the firm to choose δ. Specifically, the firm can choose $\delta \in [\underline{\delta}, \overline{\delta}]$, where $\underline{\delta}$ is the minimum level of search informativeness that the firm can choose (e.g., consumers may also search product information on other channels such as third-party reports and product reviews sites, $\underline{\delta}$ is the search informativeness on these other channels that are not controlled by the firm), and $\overline{\delta}$ is the upper bound the firm can increase the informativeness to. If the firm chooses $\delta = \underline{\delta}$, then it means that the firm does not provide extra product information and the consumer can only search information through other channels. However, if the firm chooses $\delta > \underline{\delta}$, then it means that the firm provides further information to

increase the search informativeness for the consumer. We assume for simplicity that there is no cost to the firm of providing information. We provide an outline of the solution of this model in Section A.2.3 in the appendix and state the key result in the following proposition.

Proposition 5. *Suppose the firm can choose* $\delta \in [\underline{\delta}, \overline{\delta}]$. *If* $\underline{\delta}$ *is large enough and* ξ *and* c *are small enough, the firm increases search informativeness such that* $\delta^* = \overline{\delta}$; *otherwise, the firm decreases search informativeness such that* $\delta^* = \underline{\delta}$.

Interestingly, as shown in Proposition 5, we find that if the external search informativeness $\underline{\delta}$ is high and the search cost c is low, then the firm has the incentive to further increase the search informativeness such that $\delta = \overline{\delta}$ to induce the consumer to search, and it may also allow product returns given the return processing cost is not too high. The intuition is that a high $\underline{\delta}$ and a low search cost will induce the consumer to search product information and thus the firm has the incentives to further increase the search informativeness so that the consumer is less likely to buy an ill-matched product, and it can thus charge a higher price. Furthermore, given the return processing cost is not too high, the firm will further allow product returns so that the consumer will still buy the product even if the search happens to be inconclusive, which increases the sales.

On the other hand, if the external search informativeness $\underline{\delta}$ is low or the search cost c is high, the consumer has low motivation to search and thus it is not profitable for the firm to induce consumer search by providing information. Instead, the firm will decrease search informativeness and choose the strategy to induce the consumer to buy without search, and if the return processing cost for the firm is low, it may offer generous return policy to induce the consumer to buy without search.

The main takeaway from this extension is that, when there is a return policy, the firm may want to increase or decrease search informativeness under different conditions, and it wants to set the search informativeness to one of the possible extremes. This is because the firm wants to either minimize search effectiveness for the consumer or minimize interim consumer heterogeneity arising from potentially inconclusive search as such heterogeneity reduces the firm's ability to extract consumer surplus.

4. Discussion and Conclusions

When a consumer does not know their match with a product, they can conduct a search that could give them a better idea of the match before making the purchase decision. Alternatively, if the seller offers a return policy then, depending on the terms of the policy, the consumer may be less inclined to search before purchase because they can return an ill-matched product after purchase. Therefore, it is natural to study consumer search and firm return policy together. In this paper, we build an analytical model in which consumers can conduct costly search before purchase and a seller can determine price and return policy.

Our analysis reveals that the search cost, marginal production cost and return processing cost are important parameters that determine a firm's pricing and return policy. A key takeaway is that consumer search can be used to reduce returns and save the return processing costs for the firm, whereas return policy can also be employed to offset the negative impact of consumer search on the firm's pricing capability. We also show that consumer search and product returns (with partial refund) can coexist when search is inconclusive. Another important takeaway is that due to the price reduction needed to induce consumer search, the interplay between consumer search and return policy can be counterintuitive: the ability to search can lead to more returns, while offering a return policy can result in more search. On incorporating hassle costs of returning or restrictive clauses in a return policy (e.g., 30-day return period, purchase receipt required, etc.), we show that consumers may conduct prepurchase search even if the return policy offers a full refund. Interestingly, a less costly or more informative consumer search can lead to a more generous return policy, for example, with a higher refund or more lenient return clauses. Finally, we show that if the firm can influence the informativeness of search then it will set the informativeness to be the lowest- or the highest-possible to minimize search effectiveness or minimize interim consumer heterogeneity that arises because search is inconclusive with some probability.

We are able to explain and provide new insights on a variety of pricing and return policies that firms use, such as accepting free returns with almost no conditions (e.g., Zappos), facilitating prepurchase search to reduce consumer reliance on returns (e.g., Warby Parker), not accepting returns (e.g., low-priced stores such as Dollar Tree), and accepting returns with a partial refund or with restrictive return clauses along with a full refund (most retailers). Although it might appear from lay intuition that liberal return policies (e.g., with full refunds) are offered due to competition among firms, we show that such policies can arise in a monopolistic situation and, in fact, they are always optimal if the firm can adjust the complexity of the policy by adding various clauses to the policy. Therefore, return policies that appear quite generous to consumers may actually be reducing consumer surplus.

We propose several directions for future work. First, we model search as a one-step process; it could alternatively be modeled as a sequential process (Branco et al. 2012) with the informativeness determined by the

effort exerted in the search (Jerath and Ren 2021) and could potentially be conducted both before and after purchase (Ning and Villas-Boas 2023). Second, we model inconclusive search as search randomly providing no information or full information; we could model this on the lines of Dukes and Liu (2016) where consumers can strategically do search that reveals only partial information. Third, studying a situation with firms offering multiple products could be insightful into consumer behavior, and how firms should respond optimally. For instance, at the online shoe retailer Zappos, which has a very liberal return policy, consumers often buy multiple products with the intention to only keep one of them and return all others. Our current model does not capture this situation but can be extended to capture it. Fourth, extending our framework to a situation with multiple firms would be useful to understand how our main results and insights might be modified or qualified in a competitive scenario.

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Appendix A

A.1. Analysis for the Base Model

First, to induce the consumer to search and buy only if they find a match, the firm solves the following problem:

$$E\pi^*_{(1)} = \max_p\ q(p-\psi)$$
$$\text{s.t. } EU_{\text{s}} \geq EU_{\text{ns}} \text{ and } EU_{\text{prior}} < 0.$$

The expression $EU_{\text{prior}} < 0$ implies that $p > q + (1-q)w$, and $EU_{\text{s}} \geq EU_{\text{ns}}$ implies that $p \leq 1 - \frac{c}{q}$. Therefore, for this strategy to work, we need $1 - \frac{c}{q} > q + (1-q)w$, which implies that $c < q(1-q)(1-w)$. Given $c < q(1-q)(1-w)$, it is easy to see that the optimal price is $p^* = 1 - \frac{c}{q}$, and the optimal profit is $E\pi^* = q(1-\psi) - c$. The consumer surplus is $q(1-p^*) - c$, which is equal to zero.

Next, to induce the consumer to buy without search, the firm solves the following problem:

$$E\pi^*_{(2)} = \max_{p,\theta,\gamma}\ qp + (1-q)((1-\theta)p + \theta(\gamma - \xi)) - (\theta q + (1-\theta))\psi$$
$$\text{s.t. } EU_{\text{s}} < EU_{\text{ns}} \text{ and } EU_{\text{prior}} \geq 0 \text{ and } 0 \leq \theta\gamma \leq \theta(p-w).$$

Note that $0 \leq \theta\gamma$ and $EU_{\text{prior}} \geq 0$ implies that $0 \leq \theta\gamma \leq \frac{q}{1-q} + (1-\theta)w - (\frac{q}{1-q} + (1-\theta))p$, which further implies that $p \leq \frac{\frac{q}{1-q} + (1-\theta)w}{\frac{q}{1-q} + 1 - \theta}$. Note that $EU_{\text{s}} < EU_{\text{ns}}$ implies that $0 \leq \theta\gamma < (1-\theta)(w-p) + \frac{c}{(1-q)}$, which further implies $p \leq w + \frac{c}{(1-\theta)(1-q)}$. When the firm allows product return (i.e., $\theta = 1$), it is easy to show that any p^* and γ^* that satisfy $\gamma^* = \frac{q}{1-q}(1-p^*)$ and $p^* > 1 - \frac{c}{q}$ would be the optimal solutions. We can show that if there exists an arbitrarily small size of informed consumers who already know the product match without search, then there exists only one unique solution: $p^* = 1$ and $\gamma^* = 0$. Therefore, for the ease of exposition, we select $p^* = 1$ and $\gamma^* = 0$ as the optimal solutions for further discussion in the paper. The firm obtains the optimal profit $E\pi^*_{(2),\theta=1} = q(1-\psi) - (1-q)\xi$. In this case, the consumer surplus is zero.

When the firm does not allow product return (i.e., $\theta = 0$), then if $c < q(1-q)(1-w)$, the optimal price $p^*_{(2)} = \frac{c}{1-q} + w$ and the firm can obtain profit $E\pi^*_{(2),\theta=0} = \frac{c}{1-q} + w - \psi$, and the consumer surplus is $q(1-w) - \frac{c}{1-q}$, which is higher than zero (note that $c < q(1-q)(1-w)$); if $c > \delta q(1-q)(1-w)$, then $p^*_{(2)} = q + (1-q)w$ and the firm can obtain profit $E\pi^*_{(2),\theta=0} = q + (1-q)w - \psi$, and the consumer surplus is zero.

Finally, we characterize the conditions for each strategy to be equilibrium. For the first strategy to be optimal for the firm, the parameters should satisfy $E\pi^*_{(1)} > E\pi^*_{(2)}$. Note that $E\pi^*_{(1)} > E\pi^*_{(2)}$ implies that $q(1-\psi) - c > q(1-\psi) - (1-q)\xi$ and $q(1-\psi) - c > \min\{q + (1-q)w - \psi, \frac{c}{1-q} + w - \psi\}$. These two conditions imply that the return processing cost needs to be high, and the search cost needs to be low. For the second strategy, if offering returns (i.e., $\theta = 1$) is optimal, then the parameters should satisfy $q(1-\psi) - (1-q)\xi \geq q(1-\psi) - c$ and $q(1-\psi) - (1-q)\xi \geq \min\{q + (1-q)w - \psi, \frac{c}{1-q} + w - \psi\}$. These conditions suggest that the return processing cost needs to be not high, and the search cost needs to be not too low (and also not too high if $\xi > \psi - w$). If not offering returns (i.e., $\theta = 0$) is optimal, then the parameters should satisfy $\min\{q + (1-q)w - \psi, \frac{c}{1-q} + w - \psi\} \geq q(1-\psi) - c$ and $\min\{q + (1-q)w - \psi, \frac{c}{1-q} + w - \psi\} \geq q(1-\psi) - (1-q)\xi$. These conditions suggest that the return processing cost needs to be high and the search cost needs to be high. From the analysis above, we can see that the equilibrium conditions are exhaustive and nonoverlapping.

A.1.1. Analysis for Benchmark with Search Exogenously Shut Down. For this benchmark, we exogenously shut down consumer search and the firm decides the price and the return policy. The solution of this case corresponds to the solution of the full model for large enough search cost c, specifically, $c > q(1-q)(1-w)$. The equilibria that obtain are specified in Table A.1 and illustrated in Figure A.1. We find that, when consumer search is exogenously shut down, the following outcomes obtain. If the return processing cost is high, the firm does not allow product returns, and induces the consumer to buy by charging a low price (Region (a) in Figure A.1). If the return processing cost is low, the firm charges a high price and induces the consumer to buy by allowing free product returns (Region (b) in Figure A.1). Note that Regions (a) and (b) in Figure A.1 correspond to Regions 1(a) and 1(b), respectively, in Figure 1.

Next, to understand how consumer search affects the return policy design and the firm's profit, we compare the equilibrium results in the base model with those in the benchmark, as shown in Figure A.2. There are eight regions in Figure A.2, and we discuss them one by one. Before that, we note that in the regions shared dark gray (Regions I and II) the firm makes higher profit with search than in the benchmark scenario, in the regions shaded light gray (Regions III, IV, and V) the firm makes lower

Table A.1. Equilibrium Outcomes When Consumer Search Is Exogenously Shut Down in the Base Model

Region in Figure A.1 and parametric condition	Firm strategy and induced consumer response	Expected firm profit and expected consumer surplus
Region (a)		
Return processing cost is high ($\xi > \psi - w$).	Firm does not accept product returns Induces consumer to buy by charging a low price ($p^* = q + (1-q)w$).	$q + (1-q)w - \psi$ and 0
Region (b)		
Return processing cost is low ($\xi \le \psi - w$).	Firm offers free returns Induces consumer to buy at the highest-possible price ($p^* = 1$).	$q(1-\psi) - (1-q)\xi$ and 0

profit with search than in the benchmark scenario, and in the unshaded regions (Regions VI, VII, and VIII) the firm makes the same profit with search as in the benchmark scenario.

As mentioned earlier, we find that compared with the benchmark, consumer search improves the firm's profit in Regions I and II. First, note that these regions exist only if the firm's marginal product cost is large enough, specifically, $\psi > w$. Under this condition, the firm's profit is higher in the base model compared with the benchmark if the consumer's search cost (c) is small and the firm's return processing cost (ξ) is relatively high (Regions I and II in Figure A.2). First, a high marginal production cost motivates the firm to charge a high price. Therefore, to charge a high price, the firm can either allow product returns or induce the consumer to search and resolve match uncertainty before purchasing. In Region I, the firm induces the consumer to search without returns in the base model, whereas it allows product returns (to charge a high price) in the benchmark. On the one hand, consumer search resolves the match uncertainty before purchase, and thus there would be no returns and no return processing costs for the firm—essentially, the firm transfers the cost of determining a match to the consumer rather than incur the cost of a mismatched returned product. On the other hand, because of search cost, the firm has to lower the price (compared with the benchmark) to induce the consumer to search. Overall, when the search cost is low, the first effect dominates and thus the firm can benefit from consumer search.

In Region II, where the return processing cost is quite high, the firm avoids returns—It induces the consumer to search before purchasing in the base model while it induces the consumer to buy without returns in the benchmark. In this case, consumer search benefits the firm because it allows the firm to charge a higher price and obtain a higher profit margin than that in the benchmark. Overall, in both regions, the firm can obtain a higher profit than in the benchmark.

Said differently, in both Regions I and II, the consumer searches and buys only on match in the base model but, in the benchmark, product returns are allowed in Region I and not allowed in Region II. Compared with the benchmark, in Region I the firm's higher profit is coming from saving on return costs, while in Region II they are coming from higher price.

In all other regions, that is, Regions III to VIII, the firm makes (weakly) less profit under search compared with the benchmark because search cost is high and if the firm wants to induce search it has to reduce price. Following this logic, Region III is similar to Region II, except that the

Figure A.1. Equilibrium Outcomes w.r.t. Return Processing Cost (ξ) When Consumer Search Is Exogenously Shut Down in the Base Model ($w = 0.2; q = 0.5; \psi = 0.4$)

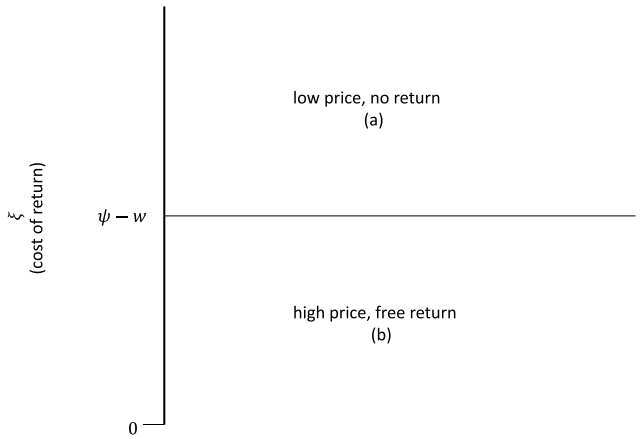
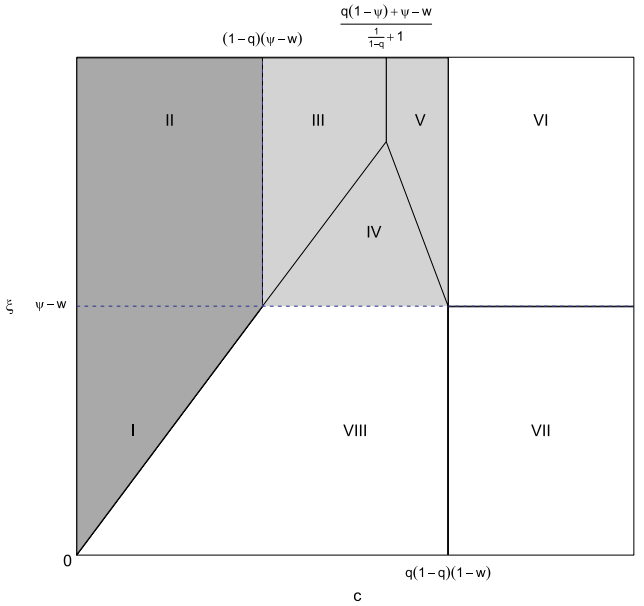


Figure A.2. (Color online) Comparison Between Base Model and the Benchmark with Consumer Search Exogenously Shut Down ($w = 0.2; q = 0.5; \psi = 0.4$)



search cost is higher in Region III, and thus the firm has to charge a lower price to induce the consumer to search (although the price is still higher than that in benchmark). Because the price is not high and the consumer buys only when they find a match through search, the firm's profit is lower than that in the benchmark. Overall, the price reduction needed to induce consumer search can be large and thus reduces the firm's profit, although search can help reduce returns. Under a different mechanism, Shulman et al. (2009) also demonstrates that consumer access to product information, despite potentially reducing returns, can concurrently decrease profits.

In Region IV, the firm does not offer returns in the benchmark, whereas it is motivated to offer product returns in the base model. That is, enabling the consumer to reduce uncertainty through search before purchase can induce the firm to implement a generous return policy, thereby increasing returns. This is in contrast to our intuition that consumer search would reduce returns. The motivation for the firm to allow product returns in this case is that the consumer search limits the firm's pricing capability and thus the firm would like to allow product returns to induce consumers to buy without search. By doing this, it can charge a high price, that is, $p^* = 1$. Note that, although the firm can charge a high price by allowing returns, the return processing cost is high and thus the firm's profit is still lower than that in the benchmark.

In Region V, the firm does not allow product returns as in the benchmark. However, because the consumer can search, the firm has to charge a very low price to induce the consumer to buy without search. This leads to a lower profit for the firm compared with the benchmark. The consumer can benefit from such a low price and obtain a positive expected surplus (in all other regions, the consumer surplus is zero.)

Finally, in Regions VI, VII, and VIII, the firm adopts the same strategy and obtains the same profit as in the benchmark.

A.1.2. Analysis for Benchmark with Returns Exogenously Shut Down. For this benchmark, we exogenously shut down returns, the consumer can search and the firm decides price. The solution of this case corresponds to the solution of the full model for large enough return processing cost ξ, specifically, $\xi > \frac{q(1-\psi)+\psi-w}{2-q}$. The equilibria that obtain are specified in Table A.2 and illustrated in Figure A.3. We find that, when returns are exogenously shut down, the following outcomes obtain. If search cost is low, the firm charges a relatively high price and induces the consumer to search and to buy only if they find a match (Region (i) in Figure A.3). If search cost is medium, the firm induces the consumer to buy without search by charging a very low price (Region (ii) in Figure A.3). If search cost is high, the firm induces the consumer to buy without search by charging a low price (Region (iii) in Figure A.3). Note that Regions (i), (ii), and (iii) in Figure A.3 correspond to Regions 2(a), 2(c), and 1(a), respectively, in Figure 1. Note that the consumer surplus is zero except when the search cost is intermediate, that is, in Region (ii).

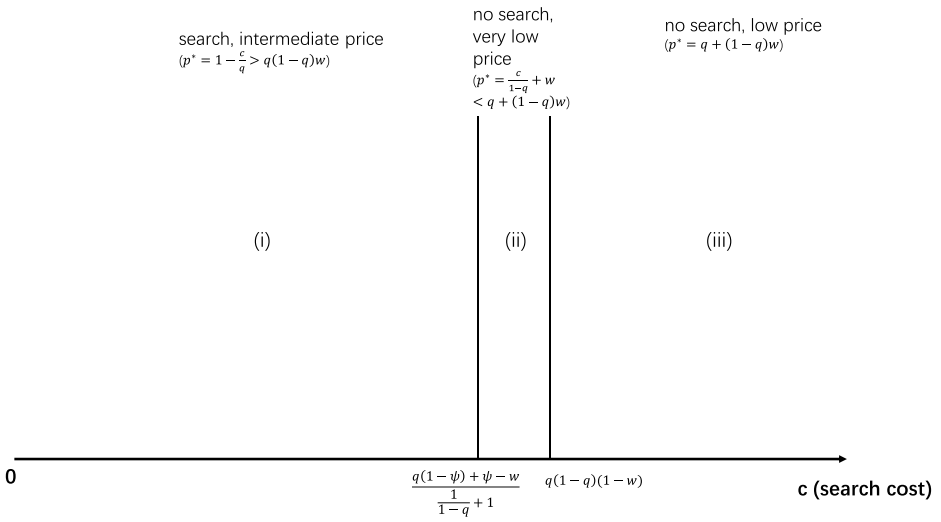
To understand how return policy affects the firm's profit and consumer surplus, we compare the equilibrium results in the base model with those in this benchmark, as shown in Figure A.4. In the unshaded regions of Figure A.4, the firm makes the same profit and adopts the same strategies in the base model as those in the benchmark. In these regions, there are no product returns. However, in the regions shaded dark gray (Regions A, B, and C), the firm allows product returns and obtains a higher profit in the base model than in the benchmark. Specifically, in Region A, the firm induces the consumer to search and resolve the match uncertainty before purchase in the benchmark, while it allows product returns and helps the consumer resolve uncertainty after purchase in the base model. Essentially, given the return processing cost is low, the firm wants to incur the cost of a mismatched returned product rather than transferring the cost of determining a match to the consumer. This allows the firm to charge a high price because it saves the consumer's search cost.

In Regions B and C, the firm induces the consumer to buy without search in the benchmark, whereas it allows product returns in the base model. In these cases, product return enables the consumer to resolve the match uncertainty after purchase, which further allows the firm to charge a high price. In particular, in Region B, the consumer can obtain a positive expected surplus without product returns in the benchmark, whereas this surplus is fully

Table A.2. Equilibrium Outcomes When Product Returns Are Exogenously Shut Down in the Base Model

Region in Figure A.3 and parametric condition	Firm strategy and induced consumer response	Expected firm profit and expected consumer surplus
Region (i) Search cost is low $\left(c \le \frac{q(1-\psi)+\psi-w}{\frac{1}{1-q}+1}\right)$.	Firm induces the consumer to search and to buy only if they find a match through search Charges a relatively high price ($p_b^* = 1 - \frac{c}{q}$; note that $p_b^* > q + (1-q)w$).	$q(1-\psi) - c\xi$ and 0
Region (ii) Search cost is medium $\left(\frac{q(1-\psi)+\psi-w}{\frac{1}{1-q}+1} < c \le q(1-q)(1-w)\right)$.	Firm induces the consumer to buy without search by charging a very low price ($p_b^* = \frac{c}{1-q} + w$; note that $p_b^* < q + (1-q)w$).	$\frac{c}{1-q} + w - \psi$ and $q(1-w) - \frac{c}{1-q} > 0$
Region (iii) Search cost is high ($c > q(1-q)(1-w)$).	Firm induces the consumer to buy without search by charging a low price ($p_b^* = q + (1-q)w$).	$q + (1-q)w - \psi$ and 0

Figure A.3. Equilibrium Outcomes w.r.t. Search Cost (c) When Product Returns Are Exogenously Shut Down in the Base Model ($w = 0.2; q = 0.5; \psi = 0.4$)



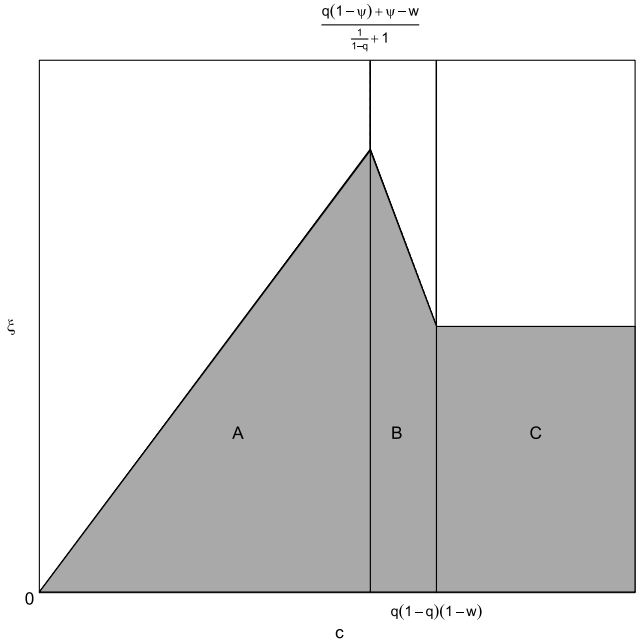
appropriated by the firm using product returns because product returns help the firm solve the price reduction needed to induce the consumer search (in other regions, the consumer surplus is always zero in both base model and the benchmark.)

A.2. Analysis for the Model with Inconclusive Search

First, to induce the consumer to search and buy only if they find a match, the firm solves the following problem:

$$E\pi^*_{(1)} = \max_p \delta q(p - \psi)$$
$$\text{s.t. } EU_s \geq EU_{ns} \text{ and } EU_{prior} < 0.$$

Figure A.4. Comparison Between Base Model and the Benchmark with Product Returns Exogenously Shut Down ($w = 0.2; q = 0.5; \psi = 0.4$)



Note that $EU_{prior} < 0$ implies that $p > q + (1-q)w$, and $EU_s \geq EU_{ns}$ implies that $p \leq 1 - \frac{c}{\delta q}$. Therefore, for this strategy to work, we need $1 - \frac{c}{\delta q} > q + (1-q)w$, which implies that $c < \delta q(1-q)\ (1-w)$. Given $c < \delta q(1-q)(1-w)$, it is easy to see that the optimal price is $p^* = 1 - \frac{c}{\delta q}$, and the optimal profit is $E\pi^* = \delta q(1-\psi) - c$. The consumer surplus is $\delta q(1-p^*) - c$, which is equal to zero.

Next, to induce the consumer to search and to purchase the product either if they find a match through search or if the search is unsuccessful, the firm solves the following problem:

$$E\pi^*_{(2)} = \max_{p,\theta,\gamma} q(p-\psi) + (1-\delta)(1-q)((1-\theta)(p-\psi) + \theta(\gamma - \xi))$$
$$\text{s.t. } EU_s \geq EU_{ns} \text{ and } EU_{prior} \geq 0 \text{ and } 0 \leq \theta\gamma \leq \theta(p-w).$$

Note that $0 \leq \theta\gamma$ and $EU_{prior} \geq 0$ implies that $0 \leq \theta\gamma \leq \frac{q}{1-q} + (1-\theta)\ w - (\frac{q}{1-q} + (1-\theta))p$, which further implies that $p \leq \frac{\frac{q}{1-q}+(1-\theta)w}{\frac{q}{1-q}+1-\theta}$. Furthermore, $EU_s \geq EU_{ns}$ implies that $\theta\gamma \geq (1-\theta)(w-p) + \frac{c}{\delta(1-q)}$. Therefore,

$$(1-\theta)(w-p) + \frac{c}{\delta(1-q)} \leq \min\left\{\theta(p-w), \frac{q}{1-q} + (1-\theta)w - \left(\frac{q}{1-q} + (1-\theta)\right)p\right\}.$$

This implies $\frac{c}{\delta(1-q)} + w \leq p \leq \min\{1 - \frac{c}{\delta q}, \frac{\frac{q}{1-q}+(1-\theta)w}{\frac{q}{1-q}+1-\theta}\}$, which further implies that $c < \delta q(1-q)(1-w)$. Given $c < \delta q(1-q)\ (1-w)$, if $\theta = 1$, the optimal price is $p^* = 1 - \frac{c}{\delta q}$ and the optimal return fee is $\gamma^* = \frac{c}{\delta(1-q)}$ and the optimal profit is $E\pi^*_{(2),\theta=1} = q(1-\psi) - c - (1-\delta)(1-q)\xi$. In this case, it is also easy to see that the consumer surplus is zero. If $\theta = 0$, $p^* = q + (1-q)w$ and the optimal profit is $E\pi^*_{(2),\theta=0} = (\delta q + 1 - \delta)\ (q + (1-q)w - \psi)$. In this case, the consumer surplus is given by $\delta q(1-p^*) + (1-\delta)q(1-p^*) + (1-\delta)(1-q)(w-p^*) - c$, which is equal to $\delta q(1-q)(1-w) - c$. This surplus is higher than zero because $c < \delta q(1-q)(1-w)$.

Finally, to induce the consumer to buy without search, the firm solves the same problem as in the base model without inconclusive search. In this case, with probability

$1-q$, the consumer will find that the product does not match, and they can return the ill-matched product if the firm allows product returns. Therefore, the firm solves the following profit maximization problem:

$$E\pi^*_{(3)} = \max_{p,\theta,\gamma} \; q(p-\psi) + (1-q)((1-\theta)(p-\psi) + \theta(\gamma-\xi))$$
$$\text{s.t. } EU_s < EU_{ns} \text{ and } EU_{prior} \geq 0 \text{ and } 0 \leq \theta\gamma \leq \theta(p-w).$$

Therefore, if the firm offers returns, $p^* = 1$ and $\gamma^* = 0$. The firm obtains the optimal profit $E\pi^*_{(3),\theta=1} = q(1-\psi) - (1-q)\xi$. If the firm does not offer returns, then $p^*_{(3)} = \min\{\frac{c}{\delta(1-q)} + w, q + (1-q)w\}$, and the firm can obtain profit $E\pi^*_{(3),\theta=0} = \min\{\frac{c}{\delta(1-q)} + w - \psi, q + (1-q)w - \psi\}$. In this case, if $c < \delta q(1-q)(1-w)$, the consumer surplus is $q(1-w) - \frac{c}{\delta(1-q)}$, which is higher than zero.

Finally, to characterize the equilibrium conditions, we compare the profits of different strategies above. For instance, for the first strategy to be optimal, we need $E\pi^*_{(1)} \geq \max\{E\pi^*_{(2),\theta=0}, E\pi^*_{(2),\theta=1}, E\pi^*_{(3),\theta=1}, E\pi^*_{(3),\theta=0}\}$. This further suggests that the return processing cost needs to be high and the search cost needs to be low. Similarly, we can characterize other conditions. Following this, we can see that there would be five parameter regions, with each of them corresponding to each strategy above, and thus the equilibrium conditions are exhaustive and nonoverlapping. The equilibrium outcomes are specified in Proposition 2 and illustrated in Figure 2.

In the outcomes, three cases (i.e., Region A, C, and E) correspond to Regions A, B, and C, respectively in Figure 1 for the base model where search is always conclusive. The other two cases in Figure 2 (Regions B and D) are new due to inconclusive search.

Region D is where return processing cost is high and search cost is intermediate. The firm does not allow product returns but charges a not-too-low price to induce the consumer to search first and buy even if search fails to provide information. The intuition is that by lowering the price (compared with Region C), the firm can increase purchases from inconclusive search. Furthermore, the consumer can obtain a positive expected surplus because the price has to be low to induce the consumer to buy when search fails and thus the consumer can obtain a positive surplus when they find a match through search (when the product matches, the consumer only needs to pay the price $p^* = q + (1-q)w$.

For Region B discussed in the main text, the proof for Corollary 3 follows by checking the equilibrium conditions for $E\pi_{(2),\theta=1}$ to be optimal.

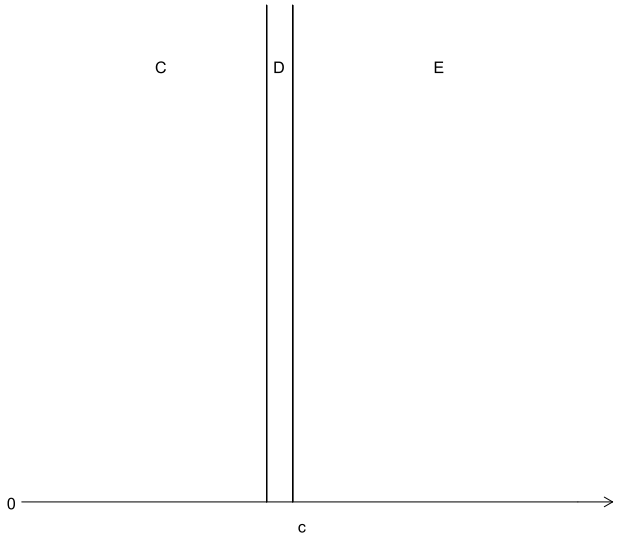
Specifically, the equilibrium conditions can be derived by

$$E\pi^*_{(2),\theta=1} > \max\{E\pi^*_{(1)}, E\pi^*_{(2),\theta=0}, E\pi^*_{(3),\theta=1}, E\pi^*_{(3),\theta=0}\}.$$

This leads to the following conditions: $\xi < \frac{q}{1-q}(1-\psi)$ and $c < \min\{\delta(1-q)\xi, \frac{q(1-\psi)+\psi-(1-\delta)(1-q)\xi-w}{1+\frac{1}{\delta(1-q)}}, q(1-\psi) - (1-\delta)(1-q)\xi - (\delta q + 1 - \delta)(q(1-\psi) + (1-q)(w-\psi))\}$. That is, the conditions suggest that the return processing cost ξ needs to be intermediate and the search cost c needs to be low.

Note that without returns (i.e., if returns are exogenously shut down), we would only have Region C for small c, Region D for medium c, and Region E for large c, and Regions A and B would not exist, as shown in Figure A.5. Therefore, contrast with the case where the return policy is shut down, subregion i of Region B of Figure 2 is the region where offering a return policy promotes consumer search, whereas subregion ii in Region A is the region where offering a return policy reduces consumer search. One can further show that subregion i exists only when $\psi < q + (1-q)w$ and $\frac{1}{2-q} < \delta < 1$. We calculate the area of this region and find that it is maximized at $\delta = \frac{\sqrt{1+4(1-q)^2}-1}{2(1-q)^2}$.

Figure A.5. Equilibrium Outcomes with Potentially Inconclusive Search w.r.t. Search Cost (c) in a Benchmark with Returns Switched Off ($w = 0.38; q = 0.5; \delta = 0.81; \psi = 0.4$)



A.2.1. Analysis with Hassle Cost. The prior utility of purchase is $EU_{prior} = q(1-p) + (1-q)((1-\theta)(w-p) + \theta(-\gamma - h))$. The expected utility of not searching is $EU_{ns} = \max\{EU_{prior}, 0\}$. The expected utility of search is

$$EU_s = \underbrace{\delta(q(1-p) + (1-q)0)}_{\text{successful search}} + \underbrace{(1-\delta)EU_{ns}}_{\text{unsuccessful search}} - c.$$

The analysis is the same as before when the firm does not offer return policy. When the firm offers returns,

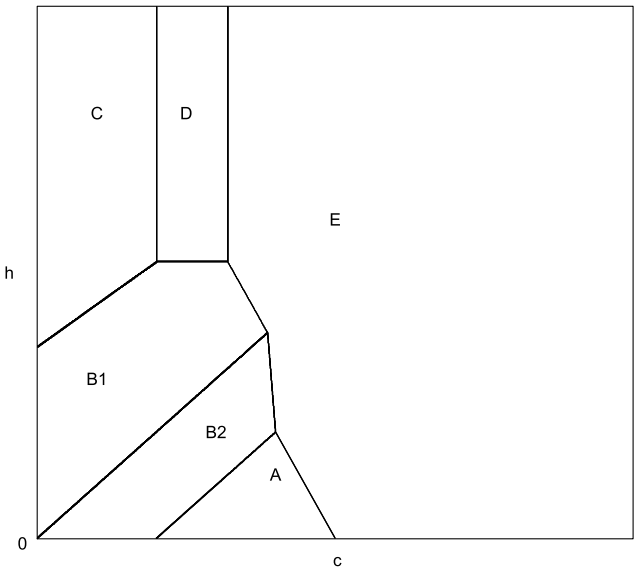
1. Consider the strategy under which consumers are induced to search and buy on a match or on an unsuccessful search, but the consumers can return the mismatched product.

The firm solves the following problem:

$$E\pi^* = \max_{p,\theta,\gamma} \; qp + (1-\delta)(1-q)(\gamma - \xi)$$
$$\text{s.t. } EU_s \geq EU_{ns}$$
$$EU_{prior} \geq 0$$
$$0 \leq \gamma \leq p - w - h.$$

The optimal solution is $p^* = \min\{1 - \frac{c}{\delta q}, 1 - \frac{1-q}{q}h\}$, $\gamma^* = \max\{\frac{c}{\delta(1-q)} - h, 0\}$. The optimal profit is $E\pi^* = \max\{q - c + (1-\delta)(1-q)(-\xi - h), q - (1-q)h - (1-\delta)(1-q)\xi\}$. In particular, when $c < \delta q(1-q)(1-w)$ and $h \leq \frac{c}{\delta(1-q)}$, we have $p^* = 1 - \frac{c}{\delta q}, \gamma^* = \frac{c}{\delta(1-q)} - h$, and $EU^* = 0$; and when $h \geq \frac{c}{\delta(1-q)}$ and $h \leq (1-w)q$, we have $p^* = 1 - \frac{1-q}{q}h, \gamma^* = 0$, and the consumer surplus $EU_s = \delta(1-q)h - c > 0$. In this case, the consumer surplus increases in h and decreases in q.

Figure A.6. Equilibrium Outcomes w.r.t. c and h ($w = 0.2$; $q = 0.5; \delta = 0.8; \xi = 0.1$)



Note. Region A corresponds to the equilibrium case: no search and generous return policy is offered ($\gamma^* = 0$); Region B1 corresponds to the equilibrium case: search and generous return policy is offered ($\gamma^* = 0$); Region B2 corresponds to the equilibrium case: search and strict return policy is offered ($\gamma^* > 0$); Region C corresponds to the equilibrium case: search and buy only if find a match and return policy is irrelevant; Region D corresponds to the equilibrium case: search and buy if find a match or if the search fails and no return policy; Region E corresponds to the equilibrium case: no search and no return policy.

2. Consider the strategy under which consumers are induced to buy without search, but the consumers can return the mismatched product.

The firm solves the following problem:

$$E\pi^* = \max_{p,\theta,\gamma} qp + (1-q)(\gamma - \xi)$$
$$\text{s.t. } EU_s < EU_{\text{ns}}$$
$$EU_{prior} \geq 0$$
$$0 \leq \gamma \leq p - w - h.$$

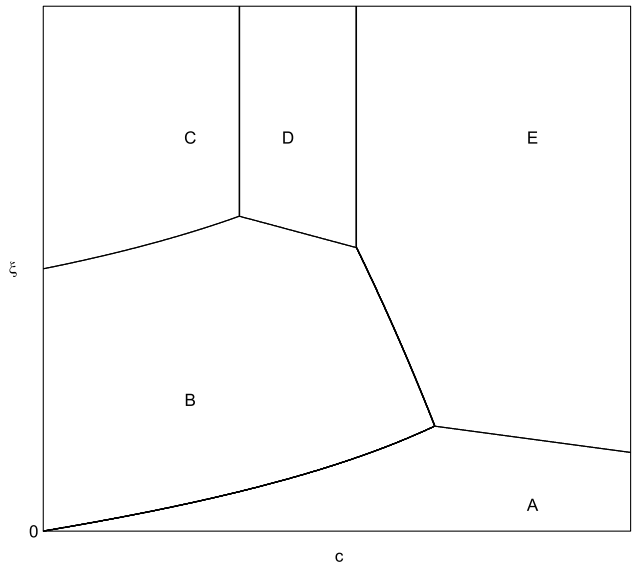
The optimal solution is $p^* = 1 - \frac{1-q}{q}h$ and $\gamma^* = 0$. The optimal profit is $E\pi^* = q + (1-q)(-\xi - h)$. The consumer obtains expected surplus $EU^* = 0$.

As in the analysis for the base model, we can then characterize the equilibrium conditions by comparing profits of different strategies. The outcomes are illustrated with the help of Figure A.6.

A.2.2. Analysis with Complexity of Return Policy. First, consider the firm's strategy to induce the consumer to search and buy if they find a match or if the search is unsuccessful, and the consumer can return the mismatched product with probability θ. The firm solves the following problem:

$$E\pi^*_{(2)} = \max_{p,\theta,\gamma} qp + (1-\delta)(1-q)((1-\theta)p + \theta(\gamma - \xi))$$
$$\text{s.t. } EU_{\text{s}} \geq EU_{\text{ns}} \text{ and } EU_{\text{prior}} \geq 0 \text{ and } 0 \leq \theta\gamma \leq \theta(p-w).$$

Figure A.7. Equilibrium Outcomes w.r.t. c and ξ ($w = 0.2$; $q = 0.5; \delta = 0.8$)



Notes. In Region A, the firm offers a generous return policy with some complexity, and the consumer does not search. In Region B, the firm offers a generous return policy with some complexity, and the consumer searches and buys on a match or on an unsuccessful search. In Region C, the firm does not offer returns and the consumer searches and buys only on a match. In Region D, the firm does not offer returns and the consumer searches and buys on a match or on an unsuccessful search. In Region E, the firm does not offer returns and the consumer buys without search.

As before, the constraints imply that $\theta\gamma \leq \min\{\theta(p-w), \frac{q}{1-q} + (1-\theta)w - (\frac{q}{1-q} + (1-\theta))p\}$ and $\frac{c}{\delta(1-q)} + w \leq p \leq \min\{1 - \frac{c}{\delta q}, \frac{\frac{q}{1-q} + (1-\theta)w}{\frac{q}{1-q} + 1 - \theta}\}$. Therefore, given $c < \delta q(1-q)(1-w)$, it is not hard to see that the optimal γ^* should satisfy $\theta\gamma^* = \frac{q}{1-q} + (1-\theta)w - (\frac{q}{1-q} + (1-\theta))p$ and the optimal price p^* should satisfy $p = \min\{1 - \frac{c}{\delta q}, \frac{\frac{q}{1-q} + (1-\theta)w}{\frac{q}{1-q} + 1 - \theta}\}$. Therefore, we can transform the above problem into the following problem:

$$E\pi^*_{(2)} = \max_{\theta} \delta q \min\left\{1 - \frac{c}{\delta q}, \frac{\frac{q}{1-q} + (1-\theta)w}{\frac{q}{1-q} + 1 - \theta}\right\} + (1-\delta)q + (1-\delta)(1-q)((1-\theta)w - \theta\xi).$$

The above problem is convex with θ and thus the optimal profit is achieved at either $\theta^* = 0$ or $\theta^* = 1 - \frac{q}{1-q}\frac{1}{\frac{\delta q(1-w)}{c} - 1}$ (such that $EU_{\text{s}} = EU_{\text{ns}}$ and $EU_{\text{prior}} = 0$). Furthermore, if $\theta^* = 1 - \frac{q}{1-q}\frac{1}{\frac{\delta q(1-w)}{c} - 1}$, it implies that $\gamma^* = 0$ because $EU_{\text{s}} = EU_{\text{ns}}$ and $EU_{\text{prior}} = 0$. In this case, θ^* decrease in c and increase in δ.

For the firm's strategy to induce the consumer to buy without search, the analysis is similar to before. Also, we can characterize the equilibrium conditions by comparing the profits of different strategies. We obtain the equilibrium outcomes as shown in Figure A.7.

A.2.3. Analysis with Endogenous Search Informativeness

Proof of Proposition 5. From the proof for Proposition 2, we can see that the firm would only have incentive to

increase the search informativeness above $\underline{\delta}$ in the following situations: (1) the firm induces the consumer to search and buy only if there is a match. This is optimal when $\underline{\delta} > \frac{c}{q(1-q)(1-w)}$ and $q-(1-q)\xi < 0$ and $c < \min\{\overline{\delta}q-(\underline{\delta}q+(1-\underline{\delta}))(q+(1-q)w), \overline{\delta}q-q+(1-q)\xi, \frac{\overline{\delta}q-w}{1+\frac{1}{\underline{\delta}(1-q)}}\}$; and (2) the firm uses return policy to induce the consumer to search and buy if there is a match or if the search is unsuccessful. This is optimal when $\underline{\delta} > \frac{c}{q(1-q)(1-w)}$ and $q-(1-q)\xi \geq 0$ and $c < \min\left\{q-(1-\overline{\delta})(1-q)\xi-(\underline{\delta}q+(1-\underline{\delta}))(q+(1-q)w), \overline{\delta}(1-q)\xi, \frac{q-(1-\overline{\delta})(1-q)\xi-w}{1+\frac{1}{\underline{\delta}(1-q)}}\right\}$. Therefore, the firm has the incentives to increase the search informativeness if and only if $\underline{\delta} > \frac{c}{q(1-q)(1-w)}$ and c is small. Furthermore, the firm would further allow product returns if $\xi < \frac{q}{1-q}$. Q.E.D.

Endnotes

[1] See https://www.shipstation.com/blog/makes-great-e-commerce-returns-policy/.

[2] See https://bit.ly/3FryjIR, https://www.newyorker.com/magazine/2023/08/21/the-hidden-cost-of-free-returns.

[3] See https://www.warbyparker.com/home-try-on.

[4] Shulman et al. (2011) also have a result that return fees in a competitive environment can be higher than those in a monopolistic environment, although their mechanism is quite different.

[5] For instance, see https://knowledge.wharton.upenn.edu/podcast/knowledge-at-wharton-podcast/high-cost-of-returns-should-retailers-rethink-policies/. This article states that: "Easy returns are great for shoppers, but they are becoming a bigger financial liability for stores."

[6] If a returned product has reduced valuation on match when being resold, even after the firm has incurred the return processing cost ξ, this can be thought of as the firm facing a salvage value of the returned product that is less than the original consumer consumption value on match of one.

[7] The return policy of Dollar Tree, available at https://www.dollartree.com/company-faq-return-policy, states: "we … do not offer refunds and consider all sales final."

[8] This is similar to Dukes and Liu (2016), where consumers can do a partial search. The main difference is that in Dukes and Liu (2016), consumers deliberately do a partial search, whereas in our case, the search may randomly fail to provide conclusive match information.

[9] Other assumptions, such as search giving a consumer only partial information about match, can also lead to search and returns coexisting because the search may change the consumer belief on fit enough for the consumer to make the purchase but the product is still found to be not a match after purchase. Moreover, we assume δ to be exogenous. In the extension in Section 3.3, we model a scenario in which the firm can endogenously set δ within limits. We also analyze a scenario in which exertion of more search effort by the consumer, which is costly to the consumer, can produce more informative search, that is, lead to larger δ. We find that our main insights do not change in this scenario. The details of this analysis are available on request.

[10] Li et al. (2019) model the offering of free samples to consumers as providing a signal that can perfectly reveal product quality with some probability and is uninformative otherwise; this is similar to our setup for consumer search.

[11] Similar to the effect of δ, the search cost c has a nonmonotonic effect on both the firm's profit and on consumer surplus.

[12] The consumer may also be able to endogenously determine the search informativeness by exerting costly effort. We find that our insights do not change qualitatively in this case.

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